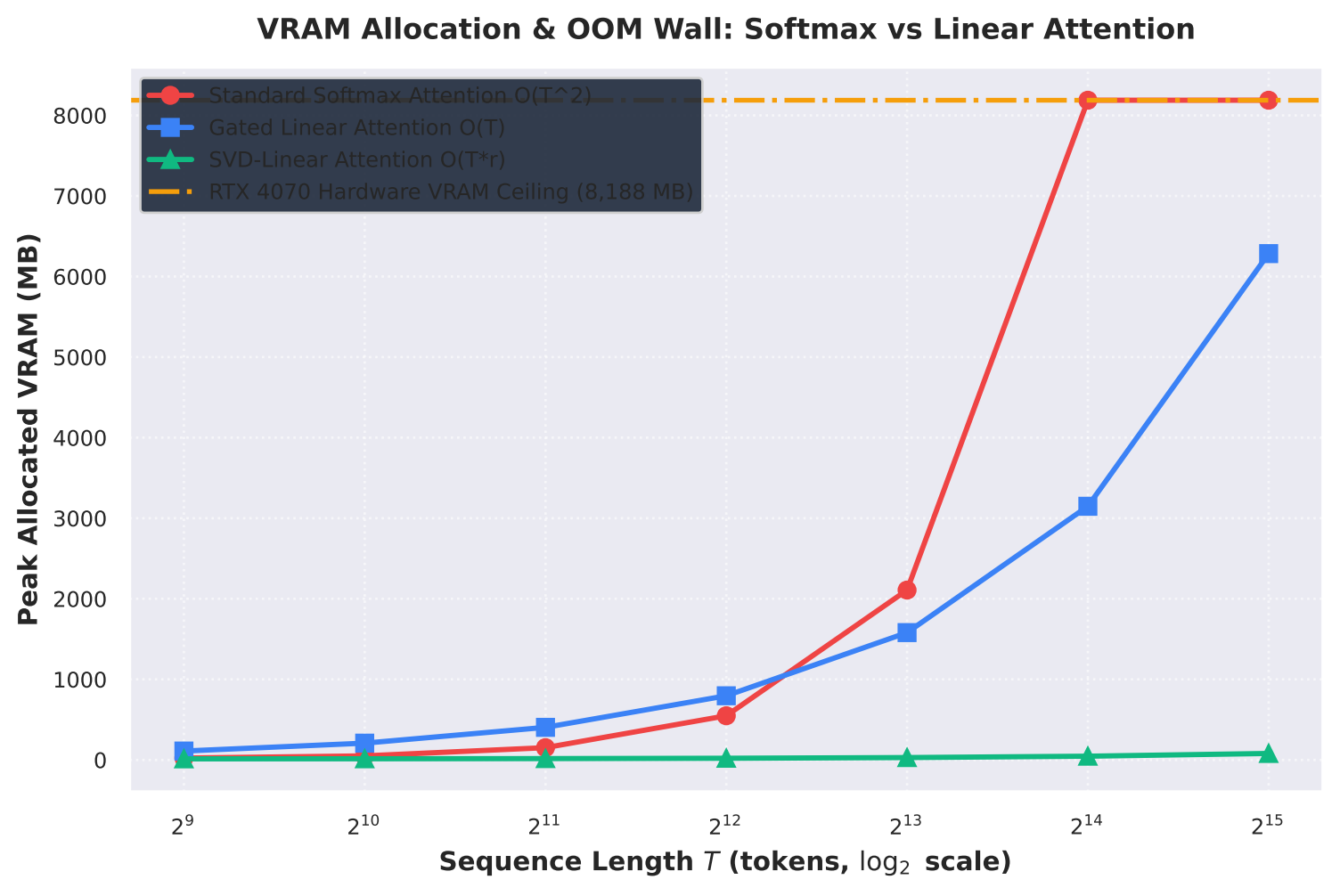
VRAM Allocation & OOM Wall: Softmax vs Linear Attention
Standard Softmax Attention O(T^2)
Gated Linear Attention O(T)
SVD-Linear Attention O(T*r)
RTX 4070 Hardware VRAM Ceiling (8,188 MB)
Peak Allocated VRAM (MB)
0
1000
2000
3000
4000
5000
6000
7000
8000
2^9
2^{10}
2^{11}
2^{12}
2^{13}
2^{14}
2^{15}
Sequence Length T (tokens, $\log_2$ scale)